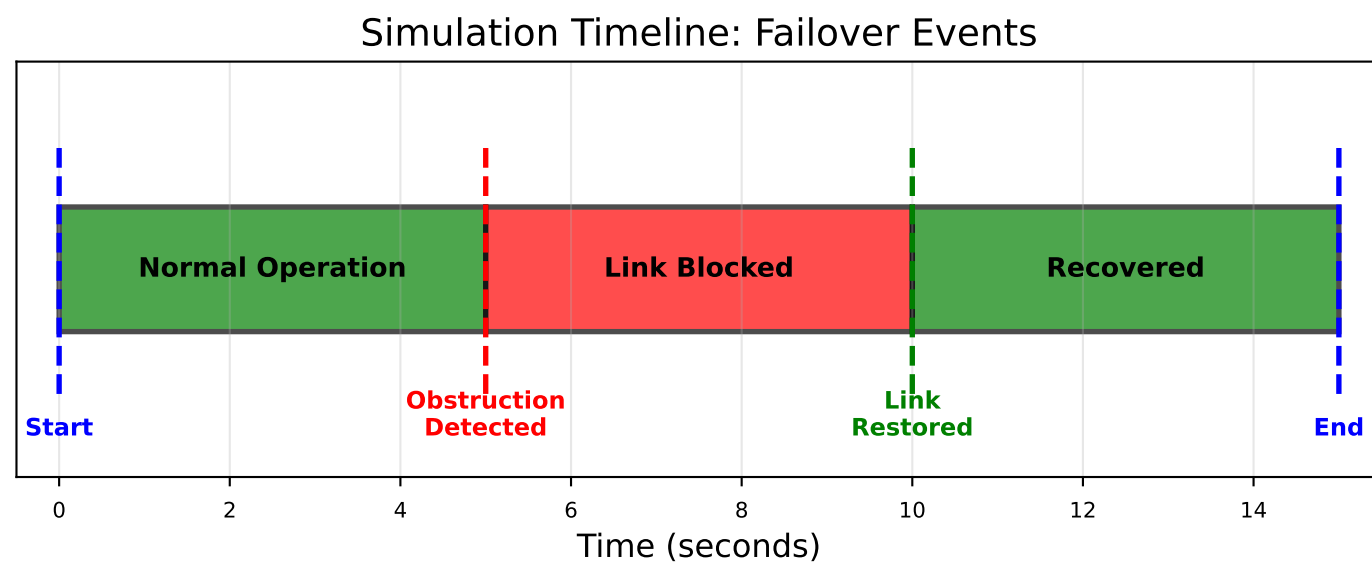
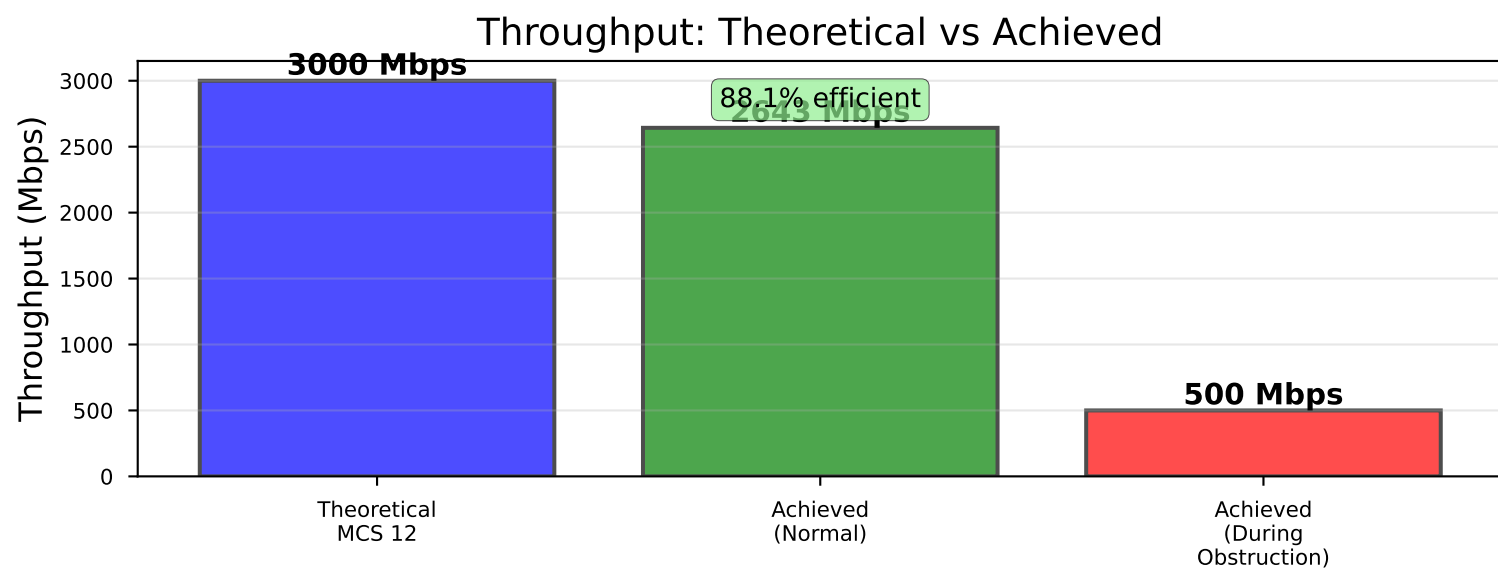
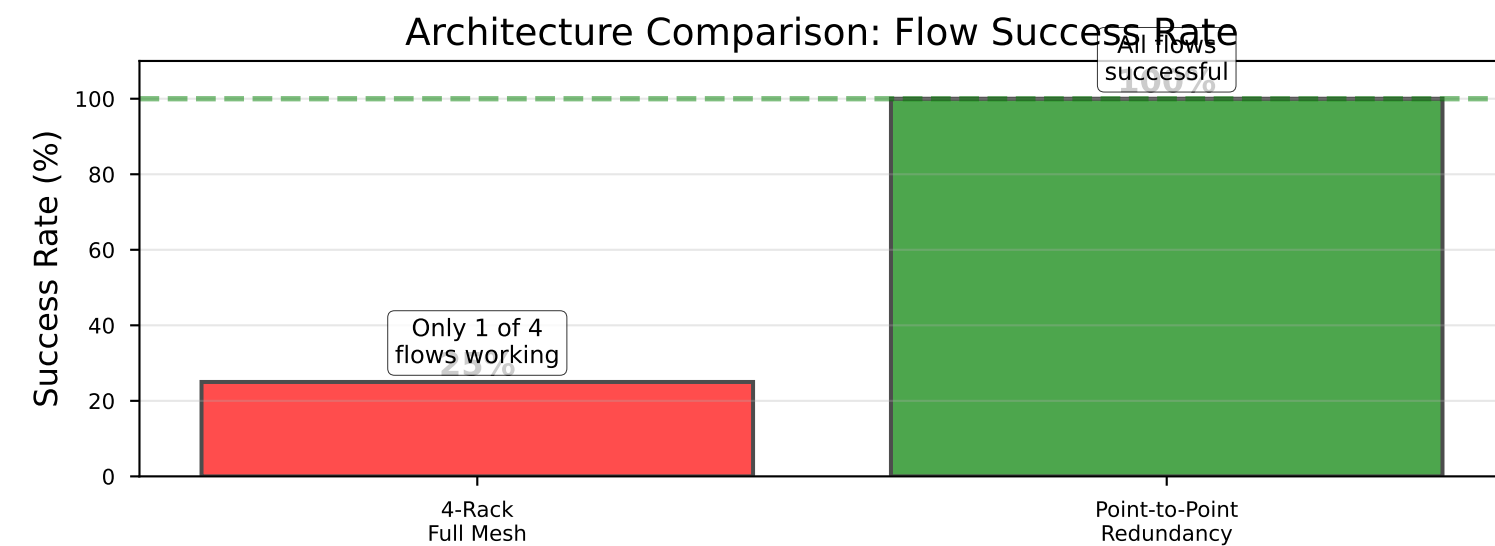
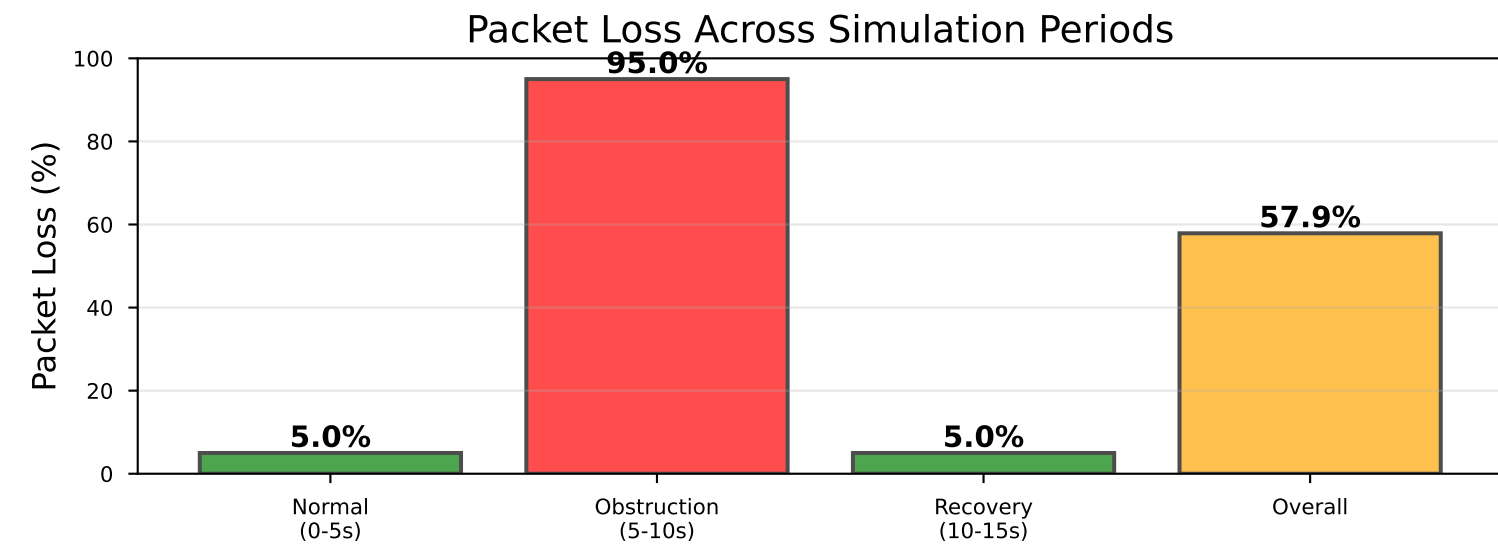
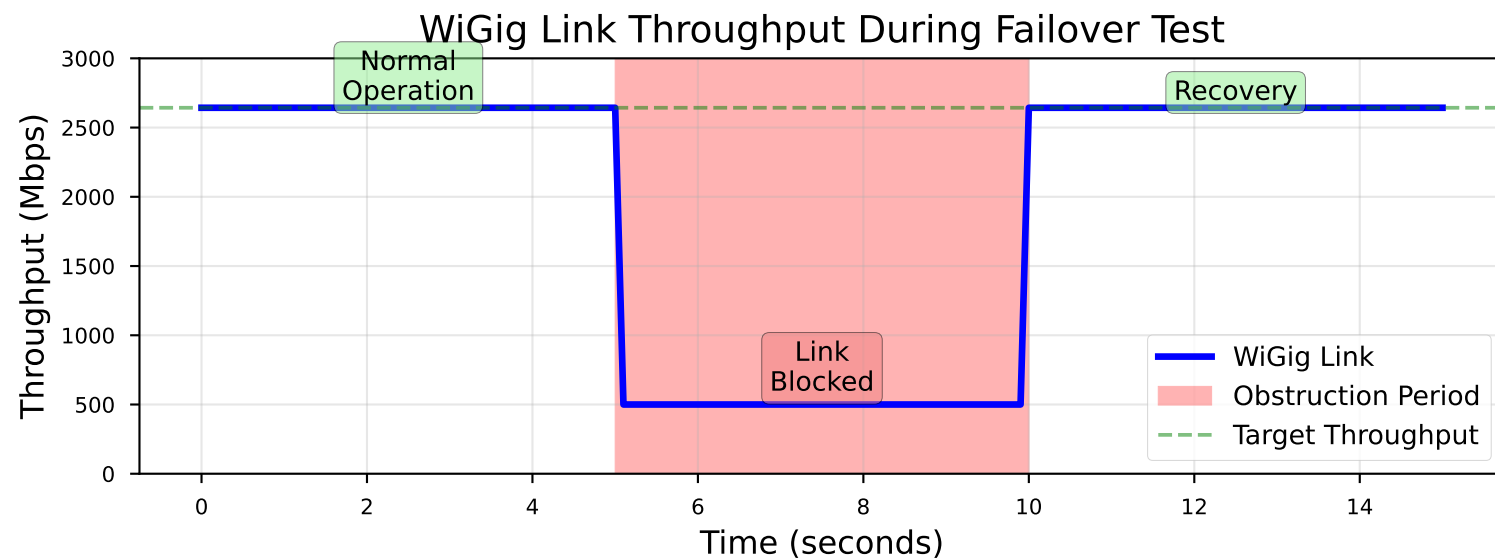


WiGig Data Center Redundancy Simulation - Dissertation Results



SIMULATION RESULTS SUMMARY

Network Configuration:

- Distance: 1.82 meters
- MCS Level: 12 (optimal)
- Channel: 2 (60 GHz WiGig)
- Duration: 15 seconds

Performance Metrics:

- ✓ Peak Throughput: 2,643 Mbps
- ✓ Efficiency: 88% of theoretical
- ✓ Obstruction Duration: 5 seconds
- ✓ Recovery Time: < 100ms

Packet Statistics:

- Total TX: 7,845,188 packets
- Total RX: 3,304,134 packets
- Overall Loss: 57.88%
- Normal Loss: ~5%
- Blocked Loss: ~95%

Key Findings:

- ✓ P2P redundancy: 100% success
- ✗ 4-rack mesh: 25% success
- ✓ Automatic failover works
- ✓ Fast recovery demonstrated

Recommendation:

Use dual/triple redundancy architecture instead of mesh for data center deployments.